

Research Methodology

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Module 1:- (Philosophy of Science)

- Objective of Science is to find out character, properties & the laws governing the things we find in nature
- Research attempts to find out what we do not know.
- Truth is Correspondence between idea, a statement, and the objective reality
- If idea, a statement accurately represents the character of something in objective reality, and predicts outcomes of an experiment conducted on matter, then idea is true
- Always be a skeptic
- Truth is Concrete, unique

- ## Subjective Thinking
- Subject thinks and whatever answer he or she arrives is true

- ## Objective
- Object tells the answer to us, by analysing, experimenting

Philosophers are broadly divided into

- Materialism - World is composed of nothing but matter
- Idealism - There are things that are supra-matter - not really matter like, :- soul, ghost etc.
- Materialism says matter is prior, idea is secondary, while idealism says vice versa
- Materialism says ~~material~~ matter exist independent of our consciousness
- While idealism says matter exists in our consciousness

- Matter is everything that exists independent of our Consciousness
- Science follow Materialism

- Mill's Causality :-

- Method of Agreement :- If 2 or more instances of phenomenon under investigation have only 1 Circumstances in Common, the Circumstances in which all alone all instances agree is cause (or effect) of given phenomenon
- Method of Difference :- If an instance in which the phenomenon under investigation occurs, and an instance in which it doesn't occur, ~~having~~ have every Circumstance save one in Common, that one occurring only in former, the Circumstance in which alone the 2 instances differ, is effect or cause, or an indispensable part of cause of phenomenon

- Joint Method of agreement and difference
Suppose 2 or more instances in which phenomenon occurs have only 1 circumstance in common. In comparison, 2 or more instances in which it doesn't occur have nothing in common except absence of that circumstance. In that case only circumstance in which 2 sets of instances differ is effect, or cause or a necessary part of cause of phenomenon

Logical Reasoning

Inductive	Deductive
(Particular to general)	(general to particular)

Inductive logic

- Understand and study particular to general conclusion by is called Inductive logic.
- This is done by inductive leap.

Basis for inductive leap :-

- Law of uniformity of nature
- " " causality

But there is always a doubt

Inductive Inferences:

- Deterministic :- unless a contrary eg is found, we can't change any fact or something
- Probabilistic :- High degree of confidence that statement / fact is true
- If we conclude general fact / statement using small sample - Simple Enumeration
- " " " " " " " "
- using Complete population - Complete Enumeration
- Improper Induction - obtained just by observation
- Proper " - ~~Causal~~ Causal Connection is understood

- Quantitative Inductive - measuring some quantity of ~~pop~~ population using a sample

Deductive logic

- If A then B
A (true)

$\therefore B$

Modus Ponens

If A then B
B (true)

$\therefore A$

Not always True

If A then B

Not A

$\therefore$ Not B

$\downarrow$

Not always True

If A then B

Not B

$\therefore$ Not A

Modus Tollens

Probabilistic Deductive logic

If A then B with probab. P
A is true

$\therefore$ B is true with probab. P

} Modus Ponens

If A then B With Probab. P } Modus
Not B } Tollens
 $\therefore$ Not A

- Defensible Deductive reasoning - Conclusion can be logically sound based on current premises, but can be defeated if new stronger info is introduced

- Syllogistic logic - structured form of deductive reasoning that requires a specific seq. to reach conclusion

1) Major Premise - All mammals have ~~hot~~ hot blood

2) Minor " - Leopard is a mammal

3) Conclusion - " is hot-blood

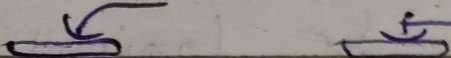
All a are b $\rightarrow$ a A b

No a is b $\rightarrow$ a E b

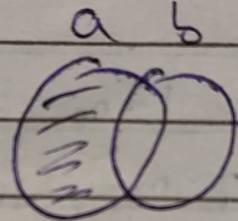
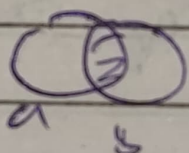
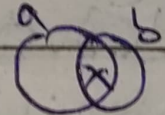
Some a are b $\rightarrow$ a I b

- a are not b $\rightarrow$ a O b

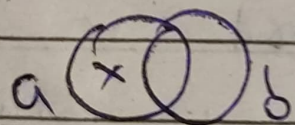
All No Some Some Not
 A E I O

Major ~~Conclusion~~ premise :- Subject Predicate
 Minor premise :- Subject SM Predicate
 Inference :- 

Venn Diagrams:-

all a are b:-  Some a are b
 No a is b:-  

Some a are not b:-



- Truth $\neq$ Validity
- The Conclusion being false implies that the premises are wrong
- The Conclusion being true doesn't imply that the premises are true